Review notes · for you

How AI is attacked and defended: unit check

AI security · Upper secondary · Unit check

Time	Materials
20 min	Paper and a pen. No device needed.

Learning goals

- Filters that miss
- Only the tools it needs
- Poisoned data
- Check before you trust a model

Activity: Defend the class bot

1. Warm-up (3 min): Recall one defence from each lesson.
2. Plan (7 min): Design the defences for a made-up class bot.
3. Attack (5 min): A partner tries four attacks on paper.
4. Share (5 min): Which defence stopped each one?

Use made-up bots, messages and files only.

Discussion

- Which defence from this unit surprised you most?
- Where is a check in code better than a rule the model reads?
- How would you explain least privilege to a younger student?

Check yourself

- My most useful defence from this unit, and where I'd use it.

Try it this week

Pick the defence from this unit you'd use most (an action check, an allow-list, the neighbour check or a fingerprint) and explain it to someone at home with a real example this week.

Quiz sheet

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Name _____ Class _____ Date _____

1 **What is a prompt injection?**

- ◯ A virus that drains the phone's battery
- ◯ A prompt so long that the model slows down
- ◯ Orders hidden in text an AI app reads, hoping the model follows them

2 **Match each attack to the defence that holds against it.**

A reworded scam message	• • • •	An allow-list per task
An agent holding ten tools	• • • •	Compare its fingerprint
Flipped labels in training data	• • • •	A check on the action in code
A model file changed on the way	• • • •	A neighbour check and majority labels

3 **What does this program print?**

```
def can_send(text, asked, yes):
    return asked and yes

print(can_send("password", True, True))
print(can_send("Hello", False, True))
```

- ◯ False, then True
- ◯ False, then False
- ◯ True, then False

4 **A reading tool can still lead to harm, even though it can’t act.**

- ◯ True
- ◯ False

5 **Which of these are data poisoning? Choose all that apply.**

- ☐ Fake reviews posted to train a shop's model
- ☐ Rain making one photo blurry
- ☐ Training photos with a planted sticker that changes the answer
- ☐ A classmate flips labels in the class dataset as a joke

6 **This allow-list check refuses the tools a task needs and allows the rest. Which line has the mistake?**

Circle the line with the mistake. Write it correctly:

```
1  def check(task, tool, yes):
2      if tool in TOOLS[task]:
3          return "not allowed"
4      if tool in ACTS and not yes:
5          return "ask first"
6      return "ok"
```

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7 **What does this program print?**

```
def suspects(xs, lab):
    out = []
    for i in range(1, len(lab) - 1):
        left, me, right = lab[i-1:i+2]
        if left != me and right != me:
            out.append(xs[i])
    return out

print(suspects([1, 2, 3, 4, 5], "SSHSS"))
```

◯ [2, 4] ◯ [3] ◯ []

8 **Put in order: using a model you downloaded.**

☐ Compute its fingerprint

☐ Download it from the official source

☐ Add it to your model list

☐ Compare it with the published one, and check the signature

9 **If a model's fingerprint matches the one posted right next to it, it must come from the real publisher.**

◯ True ◯ False

10 **Complete the sentence.**

A fingerprint proves the file has the same _____; a signature proves _____ made it.

Word bank: size · bytes · when · who

Check your answers

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1. **Orders hidden in text an AI app reads, hoping the model follows them**

   A prompt injection hides instructions in a web page, a document or a tool's result.

2. **A reworded scam message — A check on the action in code**
   An agent holding ten tools — An allow-list per task
   Flipped labels in training data — A neighbour check and majority labels
   A model file changed on the way — Compare its fingerprint

   Rewording beats word lists but not an action check; an allow-list shrinks what a tricked agent can reach; neighbour checks and majority labels expose poisoned labels; a fingerprint shows a file has changed.

3. **True, then False**

   The check ignores the text: the first message was asked for and approved, so True, even though it says "password".

4. **True**

   A reading tool can't act, but a trick can still make the agent leak or repeat what the tool read.

5. **A classmate flips labels in the class dataset as a joke; Fake reviews posted to train a shop's model; Training photos with a planted sticker that changes the answer**

   Flipped labels, fake reviews and a planted trigger (a backdoor) all put wrong examples into training data on purpose.

6. **Line 2: `if tool not in TOOLS[task]:`**

   A tool should be refused when it is not on the task's list.

7. **[3]**

   Only the H in the middle has an S on both sides, so only its value, 3, is flagged.

8. **Top to bottom: 2, 1, 4, 3**

   Start from the official source, compute the fingerprint, compare it and check the signature before you run anything, then record it so you notice if it ever changes.

9. **False**

   Whoever posted the file could post its fingerprint too.

10. **A fingerprint proves the file has the same bytes; a signature proves who made it.**

    Any change to a file changes its fingerprint, so a match means the same bytes.